

Online Book Store Report
CS-331-001
Group 4
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1. NORMALIZATION OF RELATIONS

1.1 ADMIN Relation

a) Relation and Attributes

Admin(admin_id, username, password_hash, name)

- Primary Key: admin_id
- Foreign Keys: none
- Constraints: username UNIQUE

b) Sample Data

admin_id	username	password_hash	name
1	admin01	x82hs9dh2bs	Alice Chen
2	superuser	99a82bd91kz	Mark Patel

c) Functional Dependencies

admin_id $\rightarrow$ username, password_hash, name
username $\rightarrow$ password_hash, name

d) NORMALIZATION

1NF:

- All attributes contain atomic values

2NF:

- The primary key is a single attribute.
- A single-attribute primary key cannot have *partial dependencies*, because partial dependencies only occur with composite keys.
- Every non-key attribute depends fully on admin_id.

3NF:

- We check whether any non-key attribute depends on another non-key attribute.
- username determines password_hash and name → this is acceptable because username is a *candidate key*.
- 3NF allows dependencies on candidate keys.
- There are no transitive dependencies where a non-key attribute determines another non-key attribute *and is not a key*.
- It is in 3NF

e) Decomposition Needed?

No.

1.2 CUSTOMER Relation

a) Relation and Attributes

Customer(customer_id, name, email, phone, bill_address, ship_address, registration_date)

- Primary Key: customer_id
Constraints: email UNIQUE

b) Sample Data

customer_id	name	email	phone	bill_address	ship_address	registration_date
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1	John Smit h	john@email.co m	555-1 111	10 Hill Rd	10 Hill Rd	01-NOV-2024
2	Maria Lopez	maria@email.co m	555-2 222	22 Pine St	22 Pine St	10-NOV-2024

c) Functional Dependencies

customer_id → all other attributes

email → name, phone, bill_address, ship_address

d) NORMALIZATION

1NF:

- Each attribute contains only atomic values.
- No repeating groups or multi-valued fields.

2NF:

- Primary key is the single attribute customer_id.
- No partial dependencies possible.
- All attributes depend entirely on customer_id.

3NF:

- We examine dependencies among non-key attributes.
- email is a unique candidate key → it is allowed to determine other attributes without violating 3NF.
- No non-key attribute determines another non-key attribute unless it is a key.
Customer is in **3NF**.

e) Decomposition Needed?

No.

1.3 AUTHOR Relation

a) Relation and Attributes

Author(author_id, name, nationality, biography)

- Primary Key: author_id

b) Sample Data

author_id	name	nationality	biography
1	Isaac Asimov	American	Sci-fi author
2	Haruki Murakami	Japanese	Fiction author

c) Functional Dependencies

author_id $\rightarrow$ name, nationality, biography

d) NORMALIZATION

1NF:

All values atomic.

Author is in **1NF**.

2NF:

Single primary key $\rightarrow$ no partial dependencies.

Author is in **2NF**.

3NF:

No non-key attribute determines another.

Author is in **3NF**.

e) Decomposition Needed?

No.

1.4 CATEGORY Relation

a) Relation and Attributes

Category(category_id, name)

- Primary Key: category_id

b) Sample Data

category_id	name
-------------	------

1	Fiction
---	---------

2	Science
---	---------

c) Functional Dependencies

category_id $\rightarrow$ name

d) NORMALIZATION

1NF: Atomic values

2NF: Single PK $\rightarrow$ no partial dependency

3NF: No transitive dependency

Category is in **3NF**.

e) Decomposition Needed?

No.

1.5 BOOK Relation

a) Relation and Attributes

Book(ISBN, title, edition, publication_year, price, stock_qty)

- Primary Key: ISBN

b) Sample Data

ISBN	title	edition	publication_year	price	stock_qty
9780451524	1984	1st	1949	12.99	50
9780307276	Kafka on the Shore	2nd	2002	15.50	20

c) Functional Dependencies

ISBN $\rightarrow$ title, edition, publication_year, price, stock_qty

d) NORMALIZATION

1NF: All attributes atomic

2NF: PK is single attribute $\rightarrow$ no partial dependencies

3NF: All non-key attributes depend only on ISBN and not on each other

Book is in **3NF**.

e) Decomposition Needed?

No.

1.6 BOOKAUTHOR Relation

a) Relation and Attributes

BookAuthor(author_id, ISBN)

- Primary Key: (author_id, ISBN)
- Foreign Keys:
 - author_id $\rightarrow$ Author(author_id)
 - ISBN $\rightarrow$ Book(ISBN)

b) Sample Data

author_id	ISBN
------------------	-------------

1	9780451524
---	------------

2	9780307276
---	------------

c) Functional Dependencies

$(\text{author_id}, \text{ISBN}) \rightarrow \emptyset$

d) NORMALIZATION

1NF: No multi-valued attributes

2NF: Non-key attributes? None exist. Composite key is fine.

3NF: No non-key attributes $\rightarrow$ cannot violate 3NF

BookAuthor is in **3NF**.

e) Decomposition Needed?

No.

1.7 BOOKCATEGORY Relation

Same explanation as BookAuthor.

a) Relation and Attributes

BookCategory(category_id, ISBN)

- Primary Key: (category_id, ISBN)

b) Sample Data

category_id	ISBN
--------------------	-------------

1 9780451524

2 9780307276

c) FD:

(category_id, ISBN) $\rightarrow \emptyset$

d) Normalization:

1NF: Atomic

2NF: No partial dependency

3NF: No transitive dependency

BookCategory is in **3NF**.

e) Decomposition Needed?

No.

1.8 ORDERS Relation

a) Relation and Attributes

Orders(order_id, customer_id, order_date, shipping_status, total_amount)

- Primary Key: order_id
Foreign Key: customer_id $\rightarrow$ Customer

b) Sample Data

order_id	customer_id	order_date	shipping_status	total_amount
----------	-------------	------------	-----------------	--------------

1	1	01-DEC-24	Processing	30.49
---	---	-----------	------------	-------

2	2	05-DEC-24	Shipped	15.50
---	---	-----------	---------	-------

c) Functional Dependencies

$\text{order_id} \rightarrow \text{customer_id}, \text{order_date}, \text{shipping_status}, \text{total_amount}$

d) NORMALIZATION

1NF: Atomic

2NF: Single PK

3NF: All non-key attributes depend only on order_id

Orders is in 3NF.

e) Decomposition Needed?

No.

1.9 ORDERITEM Relation

a) Relation and Attributes

OrderItem(order_id, ISBN, quantity, unit_price)

- Primary Key: (order_id, ISBN)
- Foreign Keys: order_id $\rightarrow$ Orders, ISBN $\rightarrow$ Book

b) Sample Data

order_id	ISBN	quantity	unit_price
1	9780451524	1	12.99
1	9780307276	1	15.50

c) Functional Dependencies

(order_id, ISBN) → quantity, unit_price

d) NORMALIZATION

1NF: All atomic

2NF: Both quantity and unit_price depend on full composite key

3NF: No non-key attributes determine others

OrderItem is in **3NF**.

e) Decomposition Needed?

No.

1.10 PAYMENT Relation

a) Relation and Attributes

Payment(payment_id, order_id, amount, method, payment_date)

- Primary Key: payment_id
- Foreign Key: order_id → Orders
- order_id UNIQUE

b) Sample Data

payment_id	order_id	amount	method	payment_date
------------	----------	--------	--------	--------------

1	1	30.49	Card	01-DEC-24
---	---	-------	------	-----------

2	2	15.50	PayPal	05-DEC-24
---	---	-------	--------	-----------

c) Functional Dependencies

payment_id → order_id, amount, method, payment_date

d) NORMALIZATION

1NF: Atomic
2NF: Single key
3NF: No non-key attribute determines another
Payment is in 3NF.

e) Decomposition Needed?

No.

1.11 REVIEW Relation

a) Relation and Attributes

Review(review_id, customer_id, ISBN, rating, review_date, comment_text)

- Primary Key: review_id
- Foreign Keys: customer_id → Customer, ISBN → Book
- Unique: (customer_id, ISBN)

b) Sample Data

review_id	customer_id	ISBN	rating	review_date	comment_text
1	1	9780451524	4.5	02-DEC-24	Great book!
2	2	9780307276	5.0	06-DEC-24	Masterpiece!

c) Functional Dependencies

review_id → customer_id, ISBN, rating, review_date, comment_text
(customer_id, ISBN) → rating, review_date, comment_text

d) NORMALIZATION

1NF:

Values are atomic; no multivalued attributes.

2NF:

Primary key is review_id (single attribute).

No partial dependencies.

3NF:

- Check for transitive dependencies:
(customer_id, ISBN) is a candidate key because of the UNIQUE constraint.
Dependencies from a candidate key do not violate 3NF.
- No non-key attribute determines another non-key attribute unless it is a key.

Review is in **3NF**.

e) Decomposition Needed?

No.

3. SQL Queries

Write 4 SQL Queries.

1 Contains GROUP BY

The screenshot shows the Oracle SQL Developer interface. The main window displays a SQL query in the 'Query Builder' tab:

```
SELECT author_id, COUNT(*) AS book_count
FROM BookAuthor
GROUP BY author_id;
```

Below the query, the 'Script Output' window shows the execution results. It indicates that 6 rows were inserted (though the table below shows 5 rows) and displays the following data:

AUTHOR_ID	BOOK_COUNT
1	1
2	1
3	1
4	1
5	1

1 Contains GROUP BY HAVING

The screenshot shows the Oracle SQL Developer interface. The main window displays a query in the Query Builder:

```
SELECT c_name, COUNT(o.order_id) AS num_orders
FROM Customers c
JOIN Orders o ON c.customer_id = o.customer_id
GROUP BY c_name
HAVING COUNT(o.order_id) > 0;
```

The Script Output pane shows the results of the query:

NAME	num_orders
Johnny Doe	1
Tony Stark	1
Clark Kent	1
Joe Schmo	1
Bruce Banner	1

The SQL History pane shows the execution details of the query:

SQL	Connection	Timestamp	Type	Executed	Duration/Size...
SELECT c_name, COUNT(o.order_id) AS num_orders FROM Customers c...	Fal2025	07-DEC-25 ...	SQL	1	0.033
SELECT a.name AS author_name, COUNT(b.ISBN) AS num_books FROM ...	Fal2025	07-DEC-25 ...	SQL	1	0.071
INSERT INTO ADMIN(Username, password_hash, name) VALUES (admin, ...)	Fal2025	07-DEC-25 ...	Script	3	1.076
CREATE TABLE Admin (admin_id NUMBER GENERATED BY DEFAULT AS ...)	Fal2025	07-DEC-25 ...	Script	4	0.28
DROP TABLE Review CASCADE CONSTRAINTS; DROP TABLE Payment C...	Fal2025	07-DEC-25 ...	Script	1	0.439
DELETE FROM Review; DELETE FROM Payment; DELETE FROM OrderIt...	Fal2025	07-DEC-25 ...	Script	1	0.940
SELECT category_id, COUNT(*) AS num_books FROM BookCategoryGR...	Fal2025	07-DEC-25 ...	SQL	1	0.006
SELECT author_id, COUNT(*) AS book_count FROM BookAuthorGROUP ...	Fal2025	07-DEC-25 ...	SQL	4	0.011
SELECT category_id, COUNT(*) AS num_books FROM BookCategoryGR...	Fal2025	07-DEC-25 ...	SQL	6	0.006
INSERT INTO ADMIN(Username, password_hash, name) VALUES (admin, ...)	Fal2025	07-DEC-25 ...	Script	1	2.245
CREATE TABLE Review (review_id NUMBER GENERATED BY DEFAULT ...)	Fal2025	07-DEC-25 ...	SQL	2	0.065
SELECT COUNT(*) FROM Review; SELECT COUNT(*) FROM Payment; SELE...	Fal2025	07-DEC-25 ...	Script	1	0.672
DELETE FROM Review; DELETE FROM Payment; DELETE FROM OrderIt...	Fal2025	07-DEC-25 ...	Script	1	0.241
COMMIT;	Fal2025	07-DEC-25 ...	SQL	1	0.041
DELETE FROM Book; DELETE FROM Category; DELETE FROM Author; DEL...	Fal2025	07-DEC-25 ...	Script	1	0.123
DELETE FROM Review; DELETE FROM Payment; DELETE FROM OrderIt...	Fal2025	07-DEC-25 ...	Script	1	0.184

1 Contains nested query with ALL

The screenshot shows the Oracle SQL Developer interface. The main window displays a query in the Query Builder:

```
SELECT title, price
FROM Book
WHERE price > ALL (SELECT total_amount FROM Orders);
```

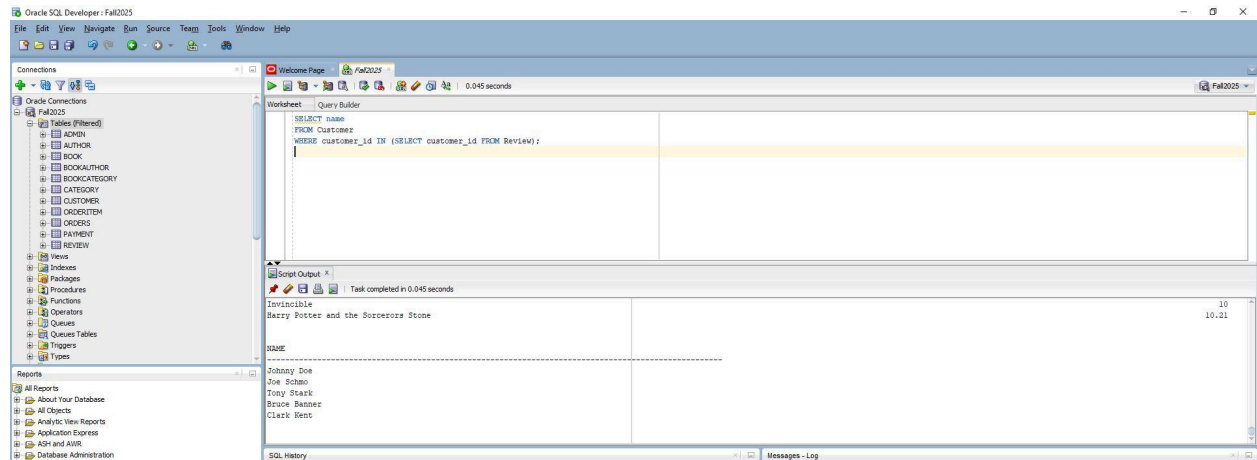
The Script Output pane shows the results of the query:

NAME
Johnny Doe
Joe Schmo
Tony Stark
Bruce Banner
Clark Kent

The SQL History pane shows the execution details of the query:

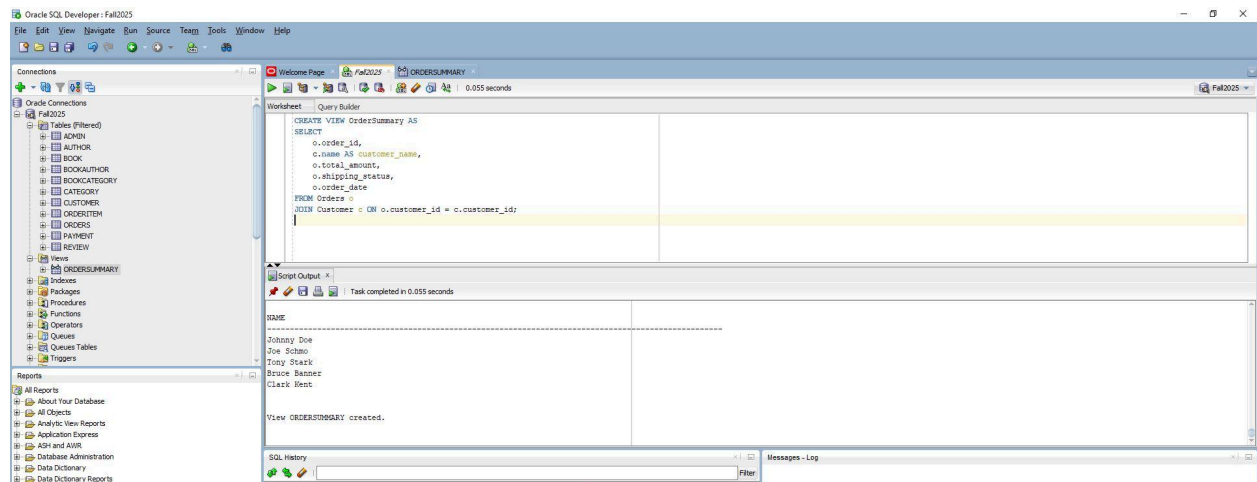
SQL	Connection	Timestamp	Type	Executed	Duration/Size...
SELECT title, price FROM Book WHERE price > ALL (SELECT total_amount FROM Orders);	Fal2025	07-DEC-25 ...	SQL	1	0.049

1 Contains nested query with IN

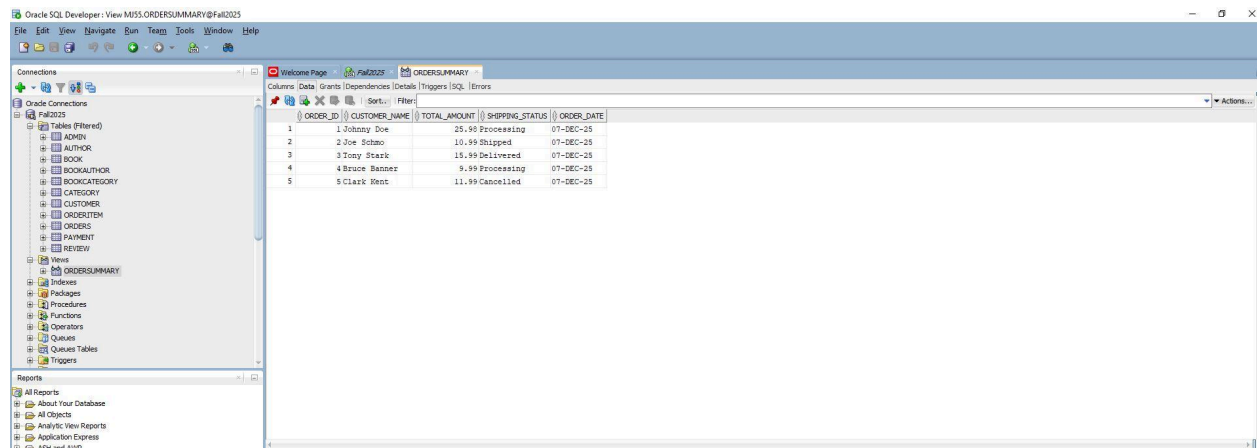


4. VIEW Query

View Code:



View Data:



5. Conclusion

This project has been very informative and fruitful for us. We as a group have learned a lot about database management and design, as well as gaining experience using SQL and SQL Developer. This project had its ups and downs, but overall we were able to learn a lot from this project and develop important skills that may help us in our careers. We collectively agreed that taking the time to learn and create proper scripts in SQL was the most difficult part of this term project. It involved knowing your database by heart and took a bit of time. It was also the most important part of this project, and the component that really helped us learn. The easiest part of this project was the normalization portion. We all agreed that it was fairly straightforward to implement and learn, even though it is also very important material. If we had to do this project again, perhaps we would manage our time a bit better. Beyond that, we believe that if we took more time to thoroughly understand the material we were working with, this project would have gone a bit smoother, but we are satisfied with our work throughout this semester.

Something else that was difficult is connecting php with the database and making sure that the data is processed correctly in the webpages. If there is a small error then it will cause issues when loading the database in the webpage. So it required us to debug and make sure that all the queries loaded on the webpage properly.